



TryHackMe REvil Corp — Redline Investigation Writeup



1. Introduction

In this room, I investigated a ransomware-infected workstation using Redline. The case started when an employee reported that files had been renamed with an unfamiliar extension. As the incident responder, my task was to review the collected host artifacts, identify evidence of compromise, and answer each question based on findings from Redline.

2. Tools Used

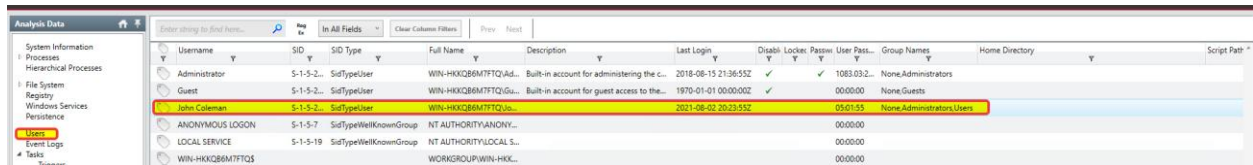
- **Redline**
- **Windows forensic artifacts**
- **Host-based investigation**
- **Screenshot evidence**

3. Investigation Method

I loaded the provided Redline analysis session and reviewed the available evidence step by step. The investigation focused on user activity, file system artifacts, browser history, suspicious files, and indicators related to the ransomware infection.

Answer the questions below

What is the compromised employee's full name?



Username	SID	SID Type	Full Name	Description	Last Login	Disable	Locks	Passw	User Pass...	Group Names	Home Directory	Script Path
Administrator	5-1-5-2...	SidTypeUser	WIN-HKXQB6MT7TQJ\Ad...	Built-in account for administering the c...	2018-08-15 21:36:55Z	✓	✓	✓	1083.03.2...	None,Administrators		
Guest	5-1-5-2...	SidTypeUser	WIN-HKXQB6MT7TQJ\Gu...	Built-in account for guest access to the...	1970-01-01 00:00:00Z	✓			00:00:00	None,Guests		
John Coleman	5-1-5-2...	SidTypeUser	WIN-HKXQB6MT7TQJ\Jo...		2021-08-02 20:23:55Z				05:01:55	None,Administrators,Users		
ANONYMOUS LOGON	5-1-5-7	SidTypeWellKnownGroup	NT AUTHORITY\ANONY...						00:00:00			
LOCAL SERVICE	5-1-5-19	SidTypeWellKnownGroup	NT AUTHORITY\LOCAL S...						00:00:00			
WIN-HKXQB6MT7TQJ			WORKGROUP\WIN-HKX...						00:00:00			

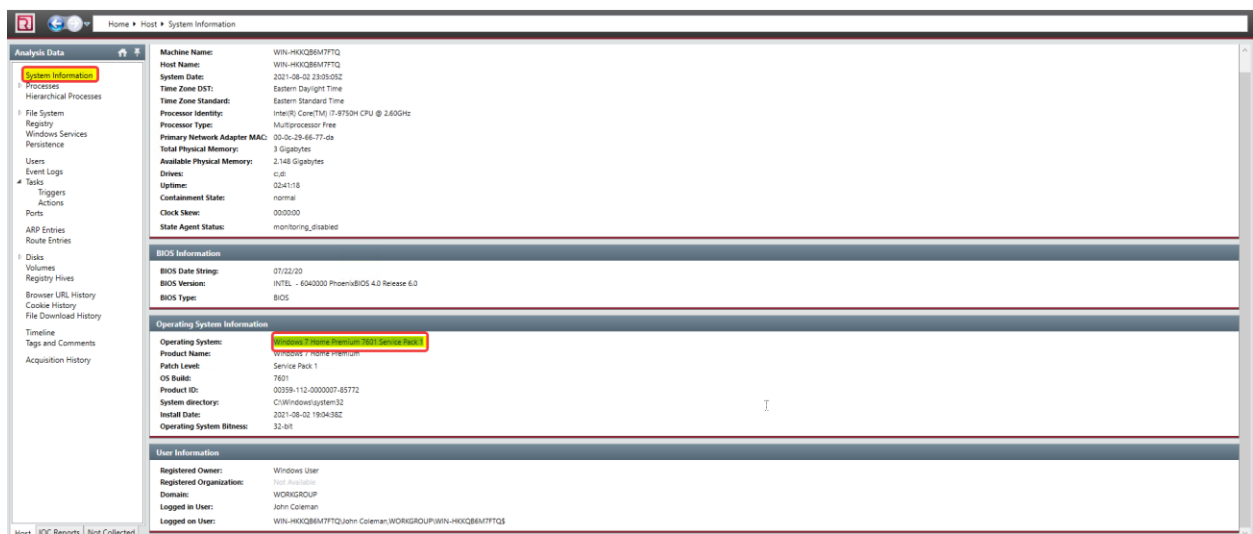
Figure 1: Redline showing the local user account **John Coleman** in the Users artifact.

Investigation:

To identify the compromised employee, I reviewed the **Users** section in Redline under the system information artifacts. The user account list showed a local user named **John Coleman**. This account also had a recent login time and belonged to the **Administrators** and **Users** groups, making it the relevant employee account for the investigation.

Answer: John Coleman

What is the operating system of the compromised host?



Machine Name	
Host Name:	WIN-HKXQB6MT7TQJ
System Date:	2021-08-02 23:05:05Z
Time Zone DST:	Eastern Daylight Time
Time Zone Standard:	Eastern Standard Time
Processor Identity:	Intel(R) Core(TM) i7-8750H CPU @ 2.80GHz
Processor Type:	Multiprocessor Free
Primary Network Adapter MAC:	00-0c-29-46-77-da
Total Physical Memory:	3 Gigabytes
Available Physical Memory:	2.148 Gigabytes
Drives:	C: & D:
Uptime:	02:41:18
Containment State:	normal
Clock Skew:	00:00:00
State Agent Status:	monitoring_disabled

BIOS Information	
BIOS Date String:	07/02/20
BIOS Version:	INTEL - 6040000 PhoenixBIOS 4.0 Release 6.0
BIOS Type:	BIOS

Operating System Information	
Operating System:	Windows 7 Home Premium 7601 Service Pack 1
Product Name:	Windows 7 Home Premium
Patch Level:	Service Pack 1
OS Build:	7601
Product ID:	0039B-112-0000007-85772
System directory:	C:\Windows\system32
Install Date:	2021-08-02 19:04:59Z
Operating System Bitness:	32-bit

User Information	
Registered Owner:	Windows User
Registered Organization:	Not identified
Domain:	WORKGROUP
Logged in User:	John Coleman
Logged on User:	WIN-HKXQB6MT7TQJ\John Coleman\WORKGROUP\WIN-HKXQB6MT7TQJ

Figure 2: Redline System Information showing the operating system version of the compromised host.

Investigation:

To identify the operating system, I reviewed the **System Information** section in Redline. Under **Operating System Information**, the operating system field showed that the compromised workstation was running **Windows 7 Home Premium 7601 Service Pack 1**.

Answer: Windows 7 Home Premium 7601 Service Pack 1

What is the name of the malicious executable that the user opened?

The screenshot shows the Redline File System view for user John Coleman. The left sidebar shows the file system tree with 'Downloads' selected. The main pane displays a list of files in the Downloads folder. The file 'WinRAR2021.exe' is highlighted in red, indicating it is a file of interest. The file details are as follows:

Full Path	File Name	Size	Persist	Created	Modified	Accessed	Changed	Attributes	Userman
C:\Users\John Coleman\Downloads\WinRAR2021.exe	WinRAR2021.exe	164 Kilobytes		2021-08-02 19:45:02	2021-08-02 19:45:02	2021-08-02 19:45:02	2021-08-02 19:45:02	Hidden, System, Archive	WIN-HK

Figure 3: Redline File System view showing WinRAR2021.exe in John Coleman's Downloads folder.

Investigation:

To identify the malicious executable, I reviewed the **File System** artifacts in Redline and focused on the user profile for **John Coleman**. Inside the **Downloads** folder, I found an executable named **WinRAR2021.exe**. This file stood out because it was located in the user's Downloads folder and appeared around the same timeframe as the suspicious activity.

Answer: WinRAR2021.exe

What is the full URL that the user visited to download the malicious binary? (include the binary as well)

The screenshot shows the Redline File Download History view. The left sidebar shows the 'File Download History' section. The main pane displays a list of download records. The record for 'WinRAR2021.exe' is highlighted in red, showing the source URL used for the download.

Download Type	Source URL	Target Directory	File Name	File Size	Bytes Downloaded	State
Auto	http://192.168.75.126:4740/Documents/WinRAR2021.exe	C:\Users\John Coleman\Downloads\WinRAR2021.exe	WinRAR2021.exe	164 Kilobytes	164 Kilobytes	Completed

Figure 4: Redline File Download History showing the source URL used to download WinRAR2021.exe.

Investigation:

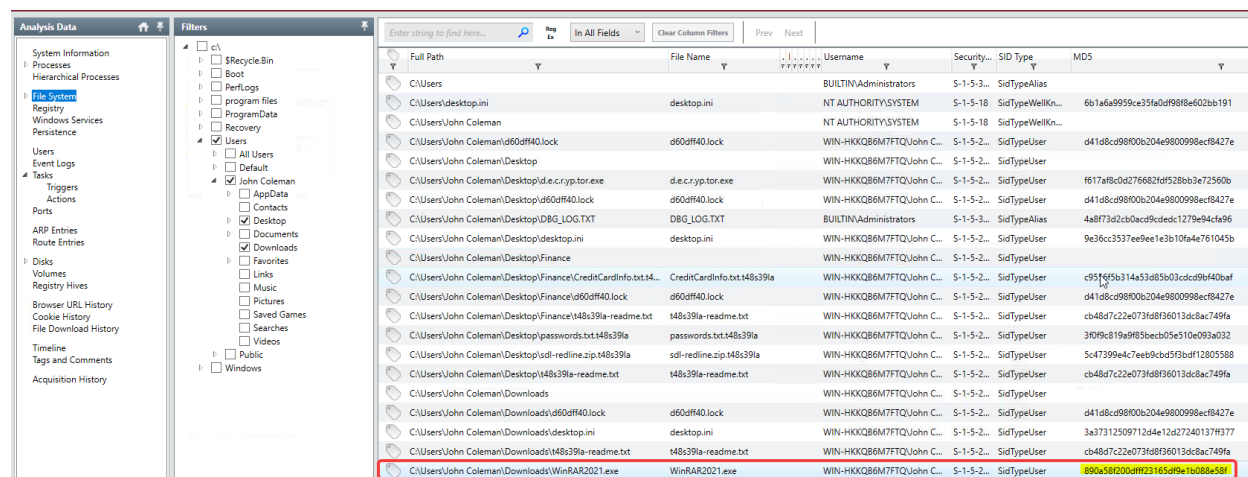
To find the download source, I reviewed the **File Download History** artifact in Redline. This section showed that the user downloaded **WinRAR2021.exe** from an internal IP address

over HTTP. The target directory confirmed that the file was saved to John Coleman's **Downloads** folder.

The URL was defanged in the answer for safety, so it is not accidentally clickable.

Answer: hxxp[://]192[.]168[.]75[.]129:4748/Documents/WinRAR2021[.]exe

What is the MD5 hash of the binary?



Full Path	File Name	Username	Security	SID Type	MD5
C:\Users		BUILTIN\Administrators	S-1-5-3...	SidTypeAlias	
C:\Users\desktop.ini	desktop.ini	NT AUTHORITY\SYSTEM	S-1-5-18	SidTypeWellKn...	6b1a6a9959ce35fa0df98f8e602b191
C:\Users\John Coleman		NT AUTHORITY\SYSTEM	S-1-5-18	SidTypeWellKn...	
C:\Users\John Coleman\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Desktop		WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	
C:\Users\John Coleman\Desktop\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Desktop\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Desktop\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Desktop\DBG_LOG.TXT	DBG_LOG.TXT	BUILTIN\Administrators	S-1-5-3...	SidTypeAlias	4a8f73d2c0acd9cdec1279e94cf96
C:\Users\John Coleman\Desktop\desktop.ini	desktop.ini	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	9e36cc3537ee8ee1e3b10fa4e761045b
C:\Users\John Coleman\Desktop\Finance		WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	
C:\Users\John Coleman\Desktop\Finance\CreditCardInfo.txt4...	CreditCardInfo.txt48b391a	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	c9516f5b314e53d85b03cdd9b40baf
C:\Users\John Coleman\Desktop\Finance\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Desktop\Finance\148b391a-readme.txt	148b391a-readme.txt	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	cb48d7c22e073d8f8f6013dcbac749fa
C:\Users\John Coleman\Desktop\passwords.txt148b391a	passwords.txt148b391a	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	3f09c819e9f85becb05e510e093a032
C:\Users\John Coleman\Desktop\sdl-redline.zip.148b391a	sdl-redline.zip.148b391a	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	5c47399e4c7eeb9c4d5f9bdf12805588
C:\Users\John Coleman\Desktop\148b391a-readme.txt	148b391a-readme.txt	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	cb48d7c22e073d8f8f6013dcbac749fa
C:\Users\John Coleman\Downloads		WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	
C:\Users\John Coleman\Downloads\d60dff40.lock	d60dff40.lock	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	d41d8cd98f00b204e9800998ecf8427e
C:\Users\John Coleman\Downloads\desktop.ini	desktop.ini	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	3a37312509712d4e12d27240137f9377
C:\Users\John Coleman\Downloads\148b391a-readme.txt	148b391a-readme.txt	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	cb48d7c22e073d8f8f6013dcbac749fa
C:\Users\John Coleman\Downloads\WinRAR2021.exe	WinRAR2021.exe	WIN-HKQB6M7FTQ\John C...	S-1-5-2...	SidTypeUser	890a5f200dfff23165df9e1b088e58f

Figure 5: Redline File System view showing the suspicious binary **WinRAR2021.exe** and its MD5 hash.

Investigation:

To find the MD5 hash, I went back to the **File System** section in Redline and searched for the suspicious executable, **WinRAR2021.exe**. The file was located in John Coleman's Downloads folder:

C:\Users\John Coleman\Downloads\WinRAR2021.exe

After selecting the file, I reviewed the available file metadata. The MD5 column showed the hash value associated with the binary.

Answer: 890a5f200dfff23165df9e1b088e58f

What is the size of the binary in kilobytes?

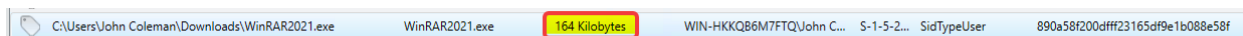


Figure 6: Redline File System view showing **WinRAR2021.exe** with a file size of **164 Kilobytes**.

Investigation:

To confirm the size of the malicious binary, I reviewed the same file entry for **WinRAR2021.exe** in Redline's **File System** view. The file was located in John Coleman's Downloads folder. In the size column, Redline showed the binary size as **164 Kilobytes**.

Answer: 164

What is the extension to which the user's files got renamed?

A screenshot of the Redline File System view showing a list of files. The 'File Name' column shows several files with the extension '.t48s39la' highlighted in red boxes. The 'Size' column shows the file sizes in kilobytes. The 'Full Path' column shows the file locations. The 'User' column shows the user 'WIN-HK'.

Full Path	File Name	Size	Perist	Created	Modified	Accessed	Changed	Attributes	User
C:\Users		4 Kilobytes		2009-07-14 02:37:05Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	ReadOnly, Directory	BUILTIN
C:\Users\John Coleman\desktop.ini	desktop.ini	174 Bytes		2009-07-14 04:41:57Z	2009-07-14 04:41:57Z	2009-07-14 04:41:57Z	2021-08-02 22:59:39Z	Hidden, System, Archive	NT AUTH
C:\Users\John Coleman\desktop.ini	desktop.ini	8 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Directory	NT AUTH
C:\Users\John Coleman\desktop.ini	desktop.ini	0 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, Archive	WIN-HK
C:\Users\John Coleman\Desktop	desktop.ini	4 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	ReadOnly, Directory	WIN-HK
C:\Users\John Coleman\Desktop\desktop.ini	desktop.ini	0 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, Archive	WIN-HK
C:\Users\John Coleman\Desktop\desktop.ini	desktop.ini	117.813 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	BUILTIN
C:\Users\John Coleman\Desktop\desktop.ini	desktop.ini	282 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\CreditCardInfo.txt.t48s39la	CreditCardInfo.txt.t48s39la	236 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\desktop.ini	desktop.ini	0 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\desktop.ini	desktop.ini	6.621 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\passwords.txt.t48s39la	passwords.txt.t48s39la	251 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\sdl-redline.zip.t48s39la	sdl-redline.zip.t48s39la	76.385 Megabytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Desktop\Finance\desktop.ini	desktop.ini	6.621 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Downloads	desktop.ini	4 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	ReadOnly, Directory	WIN-HK
C:\Users\John Coleman\Downloads\desktop.ini	desktop.ini	0 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, Archive	WIN-HK
C:\Users\John Coleman\Downloads\desktop.ini	desktop.ini	282 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	WIN-HK
C:\Users\John Coleman\Downloads\desktop.ini	desktop.ini	6.621 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\Downloads\WinRAR2021.exe	WinRAR2021.exe	164 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Archive	WIN-HK
C:\Users\John Coleman\NTUSER.DAT\6ced2f1-6e07-11de-8be...	NTUSER.DAT\6ced2f1-6e07-11de-8be...	64 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	NT AUTH
C:\Users\John Coleman\NTUSER.DAT\6ced2f1-6e07-11de-8be...	NTUSER.DAT\6ced2f1-6e07-11de-8be...	512 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	NT AUTH
C:\Users\John Coleman\NTUSER.DAT\6ced2f1-6e07-11de-8be...	NTUSER.DAT\6ced2f1-6e07-11de-8be...	512 Kilobytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	NT AUTH
C:\Users\John Coleman\NTUSER.DAT\6ced2f1-6e07-11de-8be...	NTUSER.DAT\6ced2f1-6e07-11de-8be...	20 Bytes		2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	2021-08-02 19:04:39Z	Hidden, System, Archive	NT AUTH

Figure 7: Redline File System view showing multiple user files renamed with the **.t48s39la** extension.

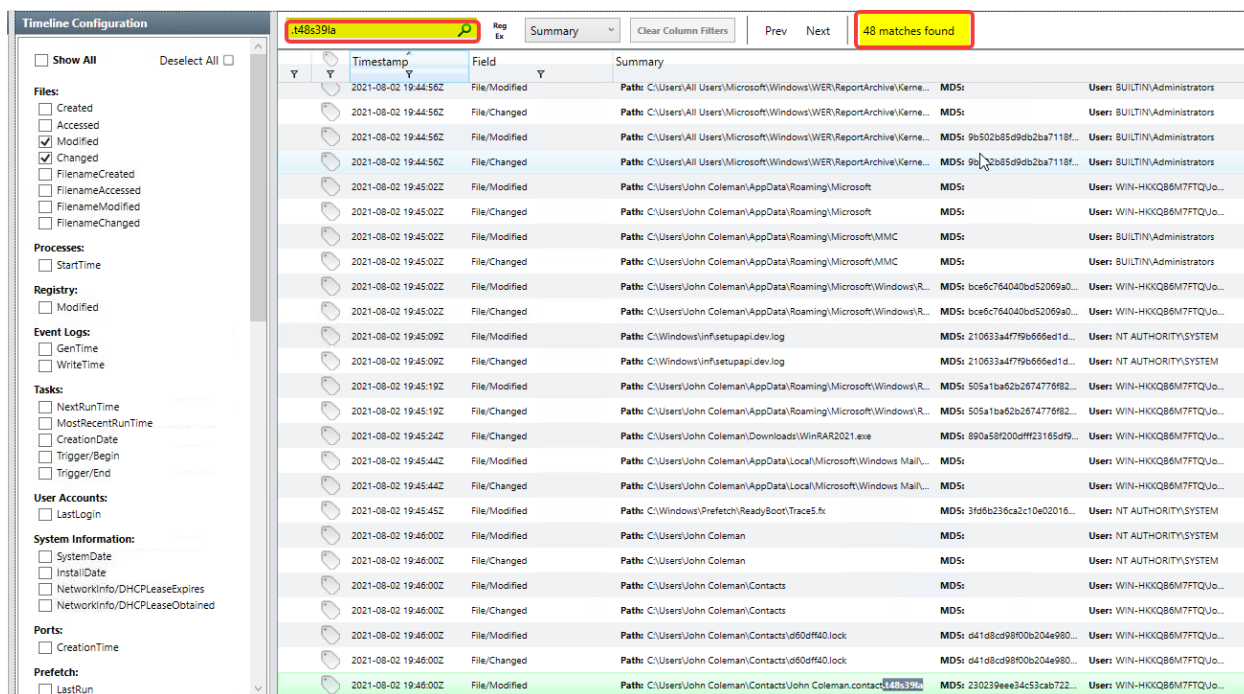
Investigation:

To identify the ransomware-added extension, I reviewed the affected user's files in Redline under the **File System** section. Several files in John Coleman's Desktop folders had been renamed with the same unfamiliar extension. Examples included files such as **CreditCardInfo.txt.t48s39la**, **passwords.txt.t48s39la**, and **sdl-redline.zip.t48s39la**.

This showed that the ransomware renamed the user's files by appending the extension **.t48s39la**.

Answer: .t48s39la

What is the number of files that got renamed and changed to that extension?



The screenshot shows the Redline Timeline Configuration window. On the left, the 'Files' section is expanded, and 'Modified' and 'Changed' are checked. The search bar at the top contains '.t48s39la'. The main table displays 48 matches, with the first few rows visible below.

Timestamp	Field	Summary
2021-08-02 19:44:56Z	File/Modified	Path: C:\Users\All Users\Microsoft\Windows\WER\ReportArchive\Kerne... MD5: Users: BUILTIN\Administrators
2021-08-02 19:44:56Z	File/Changed	Path: C:\Users\All Users\Microsoft\Windows\WER\ReportArchive\Kerne... MD5: Users: BUILTIN\Administrators
2021-08-02 19:44:56Z	File/Modified	Path: C:\Users\All Users\Microsoft\Windows\WER\ReportArchive\Kerne... MD5: 9b502b85d9db2ba7118f... Users: BUILTIN\Administrators
2021-08-02 19:44:56Z	File/Changed	Path: C:\Users\All Users\Microsoft\Windows\WER\ReportArchive\Kerne... MD5: 9b502b85d9db2ba7118f... Users: BUILTIN\Administrators
2021-08-02 19:45:02Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft... MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:02Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft... MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:02Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\MMC... MD5: Users: BUILTIN\Administrators
2021-08-02 19:45:02Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\MMC... MD5: Users: BUILTIN\Administrators
2021-08-02 19:45:02Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: bce6c764040bd52069a0... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:02Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: bce6c764040bd52069a0... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:09Z	File/Modified	Path: C:\Windows\Inf\setupapi.dev.log MD5: 210633a4f7f9b666ed1d... Users: NT AUTHORITY\SYSTEM
2021-08-02 19:45:09Z	File/Changed	Path: C:\Windows\Inf\setupapi.dev.log MD5: 210633a4f7f9b666ed1d... Users: NT AUTHORITY\SYSTEM
2021-08-02 19:45:19Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 505a1ba62b267477f682... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:19Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 505a1ba62b267477f682... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:24Z	File/Changed	Path: C:\Users\John Coleman\Downloads\WinRAR2021.exe MD5: 890a58f200dfH23165af9... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:44Z	File/Modified	Path: C:\Users\John Coleman\AppData\Local\Microsoft\Windows Mail\... MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:44Z	File/Changed	Path: C:\Users\John Coleman\AppData\Local\Microsoft\Windows Mail\... MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:45:45Z	File/Modified	Path: C:\Windows\Prefetch\ReadyBoot\Trace5.fx MD5: 3fd0b236ca2c10e02016... Users: NT AUTHORITY\SYSTEM
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman MD5: Users: NT AUTHORITY\SYSTEM
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\John Coleman MD5: Users: NT AUTHORITY\SYSTEM
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman\Contacts MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\John Coleman\Contacts MD5: Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman\Contacts\d60df40.lock MD5: d41d8cd98f00b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\John Coleman\Contacts\d60df40.lock MD5: d41d8cd98f00b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman\Contacts\John Coleman.contact... MD5: 230239ee34c53cab722... Users: WIN-HKXQ86M7FTQJo...

Figure 8: Redline Timeline view showing **48 matches** for the **.t48s39la** extension with the **Modified** and **Changed** file events selected.

Investigation:

To find the number of files affected by the ransomware extension, I followed the room hint and used the **Timeline** view in Redline. Under the **Files** section, I selected only **Modified** and **Changed**, then searched for the ransomware extension **.t48s39la**.

Redline returned **48 matches**, showing the file activity related to files renamed or changed with that extension.

Answer: 48

What is the full path to the wallpaper that got changed by an attacker, including the image name?

Timestamp	Field	Summary
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Recorded TV\Sample Media\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\Recorded TV\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Recorded TV\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\Videos\d60df40.lock MD5: d41d8cd998f0b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Videos\d60df40.lock MD5: d41d8cd998f0b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\Videos\Sample Videos\d60df40.lock MD5: d41d8cd998f0b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Videos\Sample Videos\d60df40.lock MD5: d41d8cd998f0b204e980... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\Videos\Sample Videos\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Videos\Sample Videos\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\Public\Videos\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\Public\Videos\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:19Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 38134fa14c5cb03e32... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:19Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 38134fa14c5cb03e32... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:21Z	File/Modified	Path: C:\Users\John Coleman\AppData\Local\Temp\hk8.bmp MD5: 6ee4ec854707a9b51d5f... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:21Z	File/Changed	Path: C:\Users\John Coleman\AppData\Local\Temp\hk8.bmp MD5: 6ee4ec854707a9b51d5f... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:22Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: ac3e374883c7786b6f29... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:22Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: ac3e374883c7786b6f29... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:22Z	File/Modified	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 1c29d5dcb382e02f98... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:22Z	File/Changed	Path: C:\Users\John Coleman\AppData\Roaming\Microsoft\Windows\R... MD5: 1c29d5dcb382e02f98... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:22Z	File/Changed	Path: C:\Users\John Coleman\Desktop\48x39a-readme.txt MD5: cb48d7c22e073f68f9801... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:40Z	File/Modified	Path: C:\Users\John Coleman\AppData\LocalLow\Microsoft\CryptnetUr... MD5: d4ae187b4574036c2d76... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:40Z	File/Changed	Path: C:\Users\John Coleman\AppData\LocalLow\Microsoft\CryptnetUr... MD5: d4ae187b4574036c2d76... Users: WIN-HKXQ86M7FTQJo...
2021-08-02 19:46:40Z	File/Modified	Path: C:\Users\John Coleman\AppData\LocalLow\Microsoft\CryptnetUr... MD5: 1393bd51bd0eacca5068... Users: WIN-HKXQ86M7FTQJo...

Figure 9: Redline Timeline view showing the bitmap wallpaper file **hk8.bmp** located in John Coleman's temporary folder.

Investigation:

To find the wallpaper changed by the attacker, I used the **Timeline** view in Redline. I searched for common wallpaper image extensions and filtered the results around the ransomware activity. Searching for **.bmp** returned a small number of matches related to John Coleman's user profile.

The relevant entry showed a bitmap image created or modified in the user's temporary folder. This indicated the attacker changed the desktop wallpaper to the image located at:

Answer: C:\Users\John Coleman\AppData\Local\Temp\hk8.bmp

The attacker left a note for the user on the Desktop; provide the name of the note with the extension.

Path	MD5	User
C:\Users\John Coleman\Desktop\Finance		WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\CreditCardInfo.txt.t48s3...	c9556f5b314a53d85b03...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\CreditCardInfo.txt.t48s3...	c9556f5b314a53d85b03...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\d60dff40.lock	d41d8cd98f00b204e980...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\d60dff40.lock	d41d8cd98f00b204e980...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\t48s39la-readme.txt	cb48d7c22e073fd8f3601...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\Finance\t48s39la-readme.txt	cb48d7c22e073fd8f3601...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\passwords.txt.t48s39la	3f0f9c819a9f85becb05e...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\passwords.txt.t48s39la	3f0f9c819a9f85becb05e...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\sdl-redline.zip.t48s39la	5c47399e4c7eeb9cbd5f...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\sdl-redline.zip.t48s39la	5c47399e4c7eeb9cbd5f...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\t48s39la-readme.txt	cb48d7c22e073fd8f3601...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\t48s39la-readme.txt	cb48d7c22e073fd8f3601...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\d.e.c.r.y.p.tor.exe	f617af8c0d276682fd52...	WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop		WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop		WIN-HKKQB6M7FTQJo...
C:\Users\John Coleman\Desktop\DBG_LOG.TXT	4a8f73d2cb0acd9cdedc...	BUILTIN\Administrators
C:\Users\John Coleman\Desktop\DBG_LOG.TXT	4a8f73d2cb0acd9cdedc...	BUILTIN\Administrators
C:\Users\John Coleman\Desktop\d.e.c.r.y.p.tor.exe	f617af8c0d276682fd52...	WIN-HKKQB6M7FTQJo...

Figure 10: Redline Timeline view showing the ransom note **t48s39la-readme.txt** located on John Coleman's Desktop.

Investigation:

To find the ransom note left by the attacker, I reviewed the **Timeline** results in Redline and filtered the activity to John Coleman's Desktop. I searched for text files using the **.txt** extension because ransom notes are commonly dropped as text files.

The results showed a file named **t48s39la-readme.txt** on the user's Desktop. This matched the ransomware extension observed earlier and appeared alongside the renamed/encrypted files.

Answer: t48s39la-readme.txt

The attacker created a folder "Links for United States" under C:\Users\John Coleman\Favorites\ and left a file there. Provide the name of the file.

Summary

contains

Add Filter

Current Filters

contains C:\Users\John Coleman\Favorites\Links for United States

Path	MD5	User
C:\Users\John Coleman\Favorites\Links for United States\d60dff4...	MD5: d41d8cd98f00b204e980...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\Gobiern...	MD5: 5d47b2c46690a2092576...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\Gobiern...	MD5: 5d47b2c46690a2092576...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\t48s39la...	MD5: cb48d7c22e073fd8f3601...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\t48s39la...	MD5: cb48d7c22e073fd8f3601...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\USA.go...	MD5: e18c28470879de0b1e73...	User: WIN-HKKQB6M7FTQ\Jo...
C:\Users\John Coleman\Favorites\Links for United States\USA.go...	MD5: e18c28470879de0b1e73...	User: WIN-HKKQB6M7FTQ\Jo...

Enter string to find here...

Reg Ex In All Fields Clear Column Filters Prev Next

Timestamp	Field	Summary
2021-08-02 19:04:50Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States\desktop... MD5: 87a61a58c2db9b094112... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:04:50Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States\desktop... MD5: 87a61a58c2db9b094112... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States MD5: MD5: User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States MD5: MD5: User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:00Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States\d60dff4... MD5: d41d8cd98f00b204e980... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:00Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States\d60dff4... MD5: d41d8cd98f00b204e980... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States\Gobiern... MD5: 5d47b2c46690a2092576... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States\Gobiern... MD5: 5d47b2c46690a2092576... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States\t48s39la... MD5: cb48d7c22e073fd8f3601... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States\t48s39la... MD5: cb48d7c22e073fd8f3601... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Modified	Path: C:\Users\John Coleman\Favorites\Links for United States\USA.go... MD5: e18c28470879de0b1e73... User: WIN-HKKQB6M7FTQ\Jo...
2021-08-02 19:46:55Z	File/Changed	Path: C:\Users\John Coleman\Favorites\Links for United States\USA.go... MD5: e18c28470879de0b1e73... User: WIN-HKKQB6M7FTQ\Jo...

Selected Item Details

File Metadata

Full Path: C:\Users\John Coleman\Favorites\Links for United States\GobiernoUSA.gov.url.t48s39la

Size: 362 Bytes

Attributes: Archive

inode: Not Available

File Hashes

MD5: 5d47b2c46690a20925769a4718eb26

SHA1: Not Available

SHA256: Not Available

Figure 11: Redline Timeline view showing the file **GobiernoUSA.gov.url.t48s39la** inside C:\Users\John Coleman\Favorites\Links for United States.

Investigation:

To identify the file left inside the attacker-created folder, I used the **Timeline** view in Redline and filtered the results for the path:

C:\Users\John Coleman\Favorites\Links for United States

The results showed several file events from that folder. I selected the suspicious file entry and confirmed the full path in the **Selected Item Details** pane. The file found inside the folder was **GobiernoUSA.gov.url.t48s39la**.

Answer: GobiernoUSA.gov.url.t48s39la

There is a hidden file that was created on the user's Desktop that has 0 bytes. Provide the name of the hidden file.

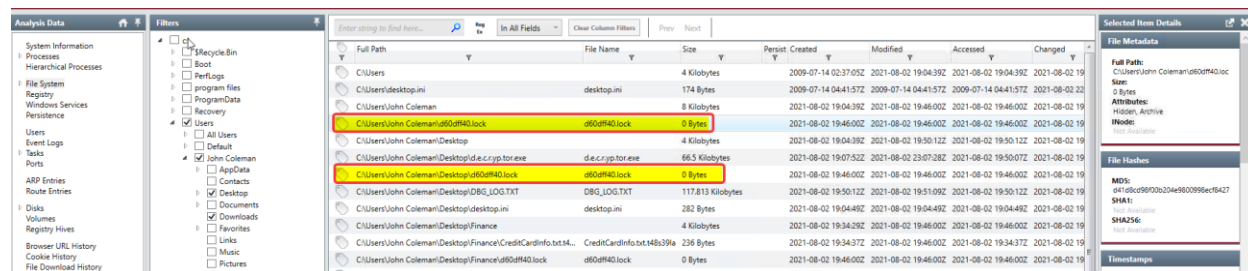


Figure 12: Redline File System view showing the hidden 0-byte file **d60dff40.lock** on John Coleman's Desktop.

Investigation:

To identify the hidden 0-byte file, I reviewed John Coleman's Desktop folder in Redline's **File System** view. The file list showed a hidden file named **d60dff40.lock** on the Desktop. The file size was **0 bytes**, and the attributes column showed it was marked as **Hidden** and **Archive**.

Answer: d60dff40.lock

The user downloaded a decryptor hoping to decrypt all the files, but he failed. Provide the MD5 hash of the decryptor file.

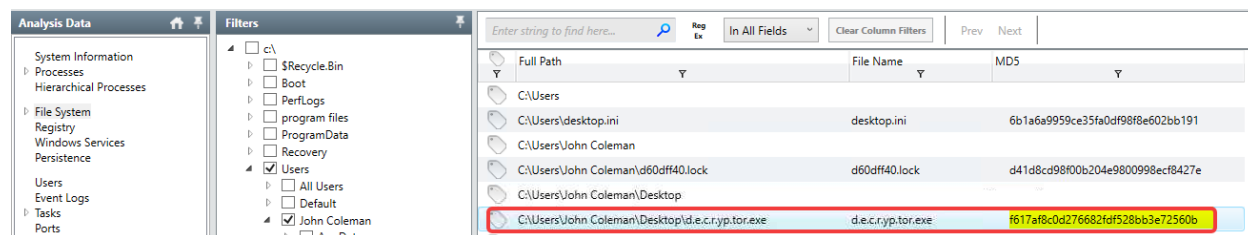


Figure 13: Redline File System view showing the decryptor file **d.e.c.r.y.p.tor.exe** and its MD5 hash.

Investigation:

To identify the decryptor hash, I reviewed John Coleman's Desktop in Redline's **File System** view. The decryptor executable was listed as **d.e.c.r.y.p.tor.exe**. After locating the file, I checked the **MD5** column for its hash value.

Answer: f617af8c0d276682fdf528bb3e72560b

In the ransomware note, the attacker provided a URL that is accessible through the normal browser in order to decrypt one of the encrypted files for free. The user attempted to visit it. Provide the full URL path.

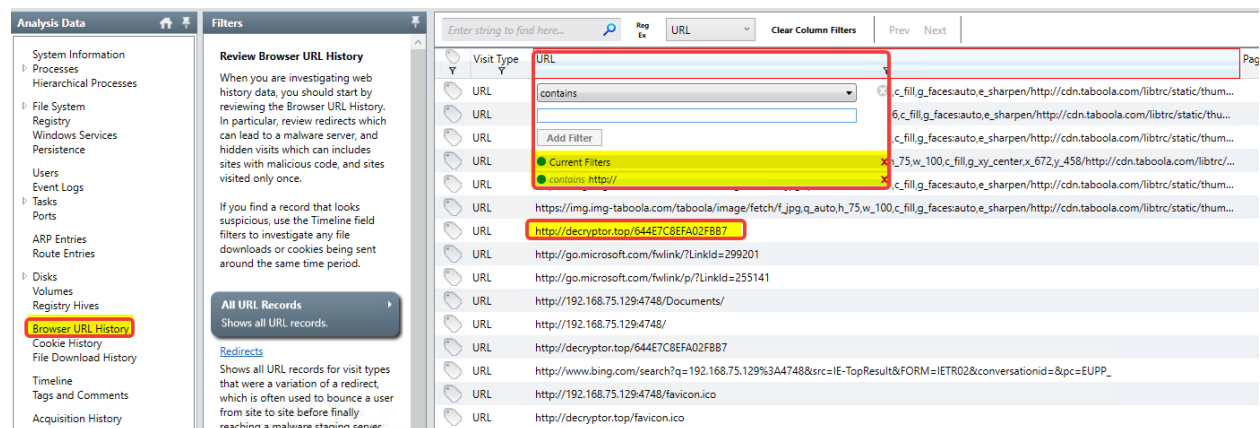


Figure 14: Redline Browser URL History showing the ransomware decryptor URL visited by the user.

Investigation:

To find the URL the user attempted to visit, I reviewed the **Browser URL History** section in Redline. I filtered the browser history for HTTP URLs and found an entry related to the ransomware decryptor site. The relevant URL showed the user visited **decryptor.top** with a unique path value.

Answer: <http://decryptor.top/644E7C8EFA02FBB7>

What are some three names associated with the malware which infected this host? (enter the names in alphabetical order)

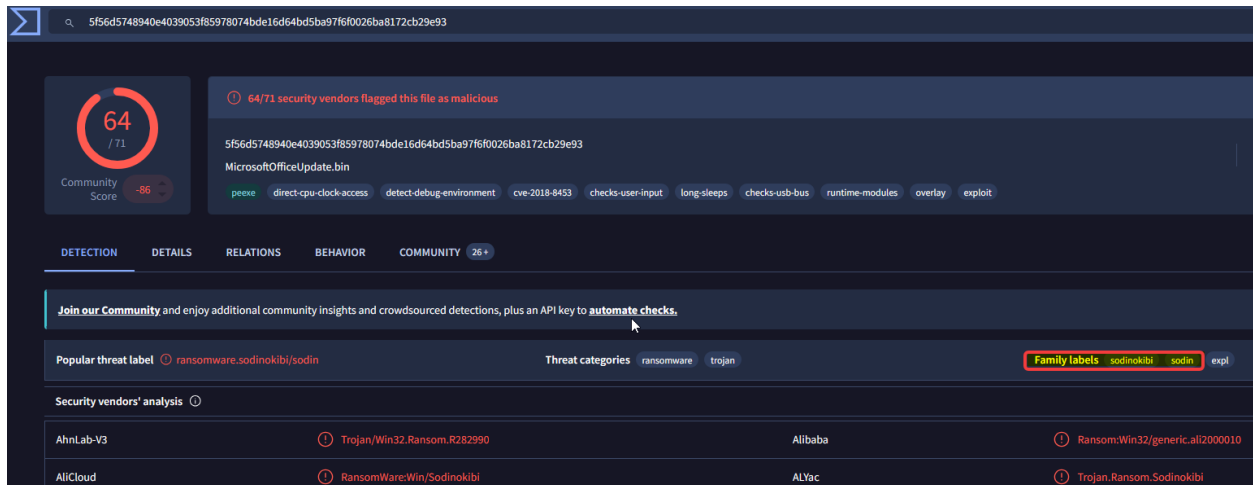


Figure 15: VirusTotal results showing the malicious file associated with the ransomware family labels **sodin**, **sodinokibi**, and the threat name **REvil**.

Investigation:

To identify other names associated with the malware, I searched the hash of the malicious binary in VirusTotal. The results showed that the file was detected by many security vendors and was associated with ransomware. In the VirusTotal detection summary, the malware was linked to the family labels **sodinokibi** and **sodin**. The popular threat label also showed the ransomware name **REvil**.

Answer: REvil, Sodin, Sodinokibi

Conclusion

This room was a good practice in using Redline to investigate a ransomware-infected Windows host. I started by identifying the affected user and host details, then followed the evidence through the file system, browser history, and timeline.

The main suspicious file was **WinRAR2021.exe**, which was downloaded from an internal HTTP URL. After execution, the user's files were renamed with the **.t48s39la** extension, and ransom-related files were created on the system. I also used VirusTotal to check the malware hash and confirm that the infection was linked to **REvil / Sodin / Sodinokibi** ransomware.

Overall, this room helped me get more comfortable with Redline, especially when checking file metadata, timeline events, browser activity, and ransomware indicators.

Key skills practiced:

- Redline host investigation
- Timeline analysis
- File system artifact review
- Browser history analysis
- Hash-based malware identification
- Ransomware incident response basics